

Task 10 - Handwritten Digit Classification **Model:** K-Nearest Neighbors (KNN) **Dataset:** Sklearn Digits (8x8 pixel images)

1. **Hyperparameter Tuning (Finding K)** We tested multiple values for **K** (the number of neighbors).

- **K=3:** Achieved the highest accuracy (~97-98%).
- **K=9:** Accuracy slightly dropped, suggesting that looking at too many neighbors introduces noise from other digits.

2. Model Performance

- **Final Accuracy:** ~97% on unseen test data.
- **Precision/Recall:** The model has near-perfect precision for digits like '0' and '6', which have very distinct shapes.
- **Confusion Matrix:** There are very few errors. Occasional confusion occurs between similar-looking digits (e.g., mistaking an '8' for a '1' if written narrowly).

3. **Conclusion** KNN is extremely effective for this dataset because handwriting styles cluster nicely in vector space. Scaling the pixel intensity features was a crucial step to ensure the Euclidean distance calculation was accurate.